

Jooho Kim

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EDUCATION

Seoul National University

BS in Electrical and Computer Engineering, Mathematical Science

Seoul, South Korea

Mar. 2021 –

- Current GPA: 3.6/4.30 ($\approx 93.0\%$)

EXPERIENCE

Republic of Korea Army — Thermal Observation Device Operator (TOD) Jul. 2022 – Sep. 2023

Mandatory Military Service

- Operated infrared (IR) thermal imaging systems for real-time surveillance and anomaly detection
- Designed and deployed a JavaScript-based scheduling system to automatically assign 20+ personnel across 12 daily shifts

RESEARCH

Undergraduate Thesis: ISI Pre-equalization for mmWave MIMO Systems

Jun. 2025 – Dec. 2025

- Modeled ultra-wideband mmWave MIMO channels and inter-symbol interference using linear signal processing theory
- Designed and implemented an ISI pre-equalization scheme in MATLAB
- Ran Monte Carlo simulations to evaluate performance under realistic multipath channels
- Improved SNR performance compared to conventional equalization baselines

PROJECTS

AI Product Recognition Competition — Encouragement Award | *Deep Learning, Python*

Jul. 2024

- Developed a deep learning-based computer vision model for automated product recognition in unmanned retail environments
- Implemented data preprocessing, augmentation, and CNN/Transformer-based architectures
- Optimized training through hyperparameter tuning and ensemble methods
- Received Encouragement Award

TECHNICAL SKILLS

Programming Languages: C/C++, Python, JavaScript, MATLAB

Developer Tools: Git, LaTeX

Languages: Korean (Native), English (Fluent)